

1. COPD: diagnosis and exacerbations

Based on NICE NG115, 2018, updated 2019, NICE NG12, 2015, updated 2025, GOLD 2025

Suspect COPD in...

Smokers/ex-smokers >35y old with:

- Exertional breathlessness or wheeze
- Chronic cough
- Regular sputum production
- Frequent 'winter bronchitis'

Features suggesting other diseases may be the cause/co-exist:

- Weight loss, fatigue, ankle swelling or breathless at night
- Occupational risk (e.g. asbestos exposure)
- Haemoptysis (if unexplained and $\geq 40y$: refer on suspected cancer pathway)
- Chest pain (urgent CXR may be offered: see NG12 for details)

Investigations

For all: • **Post-bronchodilator spirometry: $FEV_1/FVC < 0.7$ confirms diagnosis.** Reversibility not needed (because results inconsistent), but consider if diagnostic uncertainty/asthma features (see below)

GOLD takes a different approach: do pre-bronchodilator spirometry. If no obstruction ($FEV_1/FVC \geq 0.7$), COPD excluded. If obstruction is shown, proceed to post-bronchodilator FEV_1/FVC to confirm if COPD present. See main article.

- **FBC:** for eosinophil count/anaemia/polycythaemia. If thrombocytosis: think ?cancer
- **CXR:** to exclude other diagnoses
- **BMI:** obesity/cachexia

If indicated, consider:

- **If diagnostic uncertainty or asthma suspected** (symptom variability, nocturnal breathlessness/wheeze): reversibility with spirometry ($>400ml$ increase in FEV_1), serial PEF variability ($\geq 20\%$)
- **If persistent purulent sputum:** sputum culture
- **If cardiac cause suspected:** ECG/natriuretic peptide/echo
- **If other chest diseases suspected/symptoms disproportionate to spirometry:** CT chest
- **Alpha-1-antitrypsin:** young/minimal smoking/family history

Classify severity

OBJECTIVE SEVERITY

Use FEV_1 as percentage of predicted FEV_1 to assess objective severity

$\geq 80\%$	Mild/stage 1
50–79%	Moderate/stage 2
30–49%	Severe/stage 3
$< 30\%$	Very severe/stage 4

SUBJECTIVE SEVERITY

Use MRC dyspnoea score to assess subjective severity

CAUTION! There are 2 versions: MRC and mMRC
NICE uses MRC (as do our clinical systems). GOLD uses mMRC, and we've converted all the GOLD mMRC scores to MRC in the management section of this GEMS

MRC	mMRC	
1	0	Breathless only with strenuous exercise
2	1	Breathless hurrying/walking up hills
3	2	Can't keep up with peers on level ground
4	3	Stops for breath after walking 100m on flat
5	4	Breathless when dressing/housebound

SYMPTOM BURDEN

Use CAT score to assess symptom burden. A good response is a low score (< 10 considered low, maximum 40). *Link to assessment tool can be found in the main COPD articles*

COPD: treating exacerbations Based on NICE NG115, 2018

Treating acute exacerbations

- Admission needed? *Bloods/CXR/sputum culture NOT routinely needed*
- Oral steroids: **30mg prednisolone for FIVE days** (*evidence shows no benefit from longer courses*)

- Oral antibiotics: **all 5d courses**

First-line orals		Severe infection	If high risk of treatment failure (repeated antibiotic courses, high risk of complications, based on sputum culture) *Note concerns around quinolones in separate article	
Amoxicillin	500mg 3x daily	1g 3x daily	Co-amoxiclav	500/125mg 3x daily
Doxycycline	200mg on day 1 then 100mg daily	200mg daily	Levofloxacin*	500mg 1–2x daily
Clarithromycin	500mg 2x daily	-	Co-trimoxazole	960mg 2x daily (only if culture sensitive)

- If on prophylactic azithromycin, continue azithromycin but treat exacerbation with non-macrolide

Self-management if risk of exacerbations (including exacerbation in past year)

- Home rescue therapy reduces admissions (NNT 12–17/y to prevent 1 admission)
- **Start steroids if:** breathlessness interferes with activities of daily living
- **Start antibiotics if:** sputum changes colour/increases in volume/increases in thickness
- Make sure patient knows to book a review and to call for help if deteriorating on treatment

2. COPD: Managing stable disease

Based on NICE NG115, 2018 and 2019 and GOLD 2025

Health promotion

- Encourage smoking cessation and document pack years (no. cigarettes/day divided by 20 then x no. years smoked for)
- Look for and treat anxiety/depression, obesity/cachexia
- Pneumococcal/flu immunisation
- **Pulmonary rehab** if admission with exacerbation or MRC ≥ 3 (can't keep up with peers on level ground)

Inhaled drug therapy

We've summarised both NICE and GOLD guidelines, drawing out similarities and differences (mainly around who benefits from ICS – see main article for detail). We think most will move towards GOLD guidance.

Which inhaler device? Dry-powder devices and soft mists have lower carbon footprints than pMDIs; however, some dry-powder devices need significant inspiratory effort, which some patients with COPD may not have. Consider whether the patient has sufficient inspiratory effort before use: see next page for details.

Short-acting agent for prn relief (SABA preferred: SAMA can't be used with LAMA once step-up therapy needed)

More treatment needed?

Ongoing symptoms/breathlessness/frequent exacerbations (this will be the majority, we think)

Start long-acting drugs

How do I choose which long-acting drugs to use?

NOTE: we've converted the mMRC scores in the GOLD guidance to the more widely used MRC scores (see diagnosis section for more details)

GOLD calls this group B:
MRC ≥ 3
CAT ≥ 10

The 2 guidelines agree: for most, it is:

LABA + LAMA

UNLESS there is a good reason to use ICS

(i.e. the benefits of ICS outweigh the pneumonia risk)

Who should have ICS?

NICE and GOLD define these groups slightly differently as outlined below, but **both agree that those with asthma/PMH of asthma should be on ICS**

EXCEPTION! GOLD 2025 has a group which will be on a single long-acting agent (we think this will be a small group, and, under NICE, this group would be on short-acting agents alone):

FEW SYMPTOMS and FEW EXACERBATIONS

GOLD suggests:

LABA or LAMA

Defined by GOLD as:

Low symptom score: MRC < 3 , CAT < 10 and < 2 exacerbations/y and no admissions

GOLD calls this group A

NICE recommends ICS in all those with:

Features of 'steroid responsiveness'

NICE defines steroid responsiveness as:

- PMH asthma/atopy
- Symptom variability
- Higher blood eosinophils

NICE recommends ICS used with LABA

ICS + LABA

GOLD says consider ICS in those with asthma AND in those with:

FREQUENT EXACERBATIONS
(≥ 2 /y or admission in past year)

AND

BLOOD EOSINOPHILS ≥ 0.3

GOLD recommends ICS used with LABA AND LAMA

ICS + LABA + LAMA

Why straight to triple therapy? GOLD says that triple therapy is superior to ICS+LABA and should therefore be used

Remember this as the Es: Exacerbations and Eosinophils of 0.3 (point thrEE) or more

GOLD calls this group E

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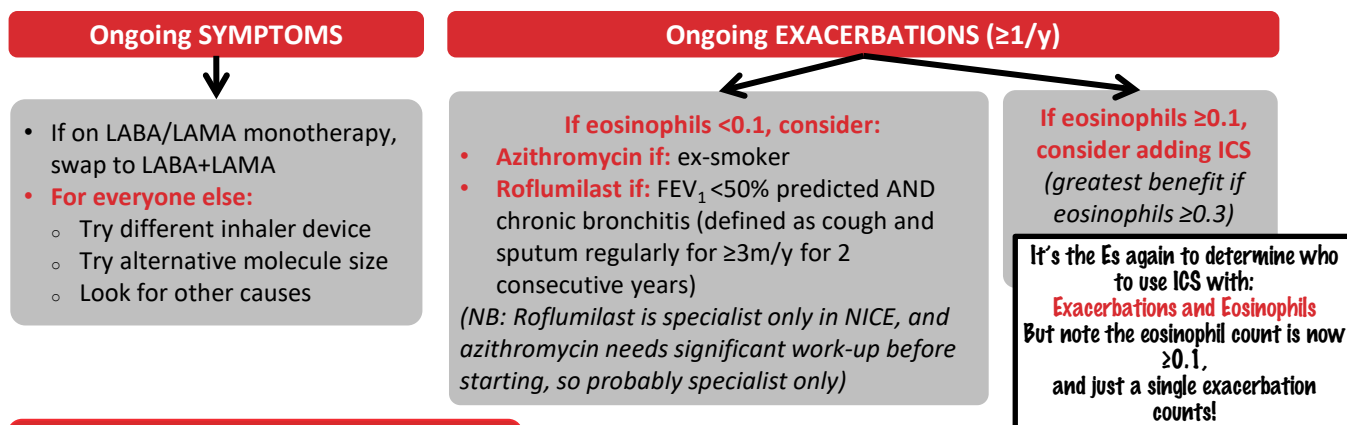
3. COPD: step-up therapy and beyond

Based on NICE NG115, 2018 and 2019 and GOLD 2025

If initial long-acting therapies are insufficient, what next?

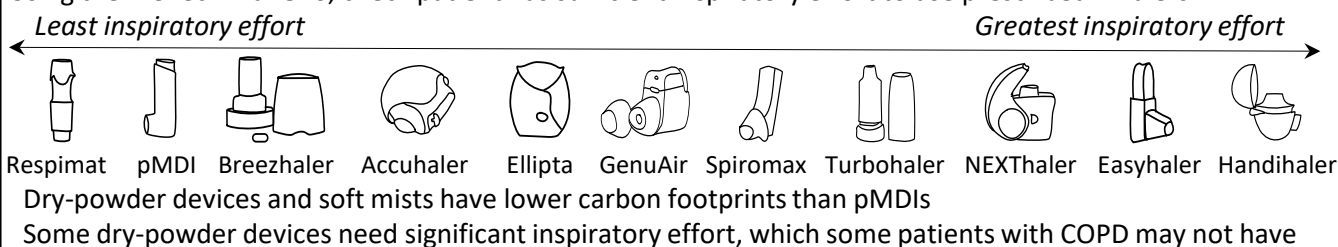
- Check adherence and ability to use device (good understanding, manual dexterity, sufficient inspiratory effort)
- Consider if comorbidities are contributing
- Consider non-pharmaceutical interventions: smoking cessation, pulmonary rehab

Here, we summarise the GOLD guidance. For those who want to follow NICE, see the main article



Inspiratory effort and inhaler choice

Using the InCheck Dial G16, check patient has sufficient inspiratory effort to use prescribed inhalers



The rest of this article is based on NICE guidance

Consider prophylactic antibiotics if:

Non-smoker and significant daily sputum, and:

Usually, azithromycin 250mg 3x a week

- Hospitalised by exacerbation, prolonged exacerbations or ≥4 exacerbations/year

Significant work-up needed first; risk of tinnitus/hearing loss and limited evidence: specialist decision?

Theophylline (slow release)

Only if inhalers insufficient. Needs plasma monitoring. Reduce dose if treated with macrolide/quinolones

Referral criteria

- Diagnostic uncertainty or symptoms disproportionate to spirometry/dysfunctional breathing
- Suspected alpha-1-antitrypsin deficiency (onset under 40y/family history)
- Signs of other disease: **haemoptysis** (cancer?) or **frequent infections** (bronchiectasis?)
- Rapid decline in FEV₁ (≥500ml loss in FEV₁ in <5y) or severe disease/develops cor pulmonale
- Assess need for: nebulisers/maintenance oral steroids/oxygen/surgery
- **Refer for long-term oxygen therapy (which must be used for ≥15h a day) if:**
 - FEV₁ <30% (consider if FEV₁ 30–49%)
 - Oxygen sats ≤92%
 - Cyanosis/polycythaemia

Palliative care

Multidisciplinary care, including hospices

If needed for breathlessness, use opioids, benzodiazepines, major tranquillizers, tricyclics and oxygen

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